



Value of time multipliers for walk time, wait time, service headway, displacement time, reliability and congested travel time: review and meta-analysis of worldwide evidence

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ABSTRACT

This study provides the most comprehensive review and meta-analysis of time multipliers that are fundamental to transport planning and appraisal. The attributes covered are: walk time, wait time, access/egress time, connection time, out-of-vehicle time and service headway, which are largely characteristics of public transport trips; travel time in various degrees of congestion and parking search time, which are features of car travel; displacement time; and a range of variables related to travel time variability. The evidence assembled covers 2872 multipliers drawn from 606 studies across 52 countries over the years 1963 to 2023.

It was found that many of the multipliers vary between Revealed and Stated Preference data, with the former almost always larger, and over time, indicating a reduction where an effect was discerned. Other influences were detected from mode used, journey purpose and distance. The estimated meta-model implies credible multipliers across a range of illustrative scenarios. It can be used to deliver multipliers for use in planning and appraisal where none exist or to benchmark existing or emerging evidence. When compared with official recommendations, the meta-model's implied multipliers challenge the long-established multipliers of two for walk and wait time, supporting a reduction to 1.75 or 1.50 depending upon the emphasis placed on Revealed or Stated Preference data. The results support official headway and late time multiplier recommendations but suggest refinements for the reliability ratio and congested time multipliers along with new guidance for displacement time multipliers.

1. Introduction

1.1. Objectives and context

Travel time is at the heart of transport planning, forecasting, policy and appraisal. The value of travel time savings (VTTs) is often

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cited as the single most important parameter in transport appraisal (Mackie et al., 2001; Mouter, 2020). But travel time is incredibly miscellaneous. It has long been recognised that there are many other types of time associated with travel in addition to in-vehicle time (IVT). These can be differentiated according to: the effort involved, of which the most notable is walking time; the conditions of experienced time, such as waiting and congested traffic; the stage of the trip, covering accessing a station or vehicle or time spent interchanging; convenience related factors, namely service frequency and the ability to depart at the desired time; and the uncertainty surrounding travel time, represented by late arrivals or some measure of the spread of journey times. The theory (DeSerpa, 1971) underpinning VTTS contains a term relating to the marginal utility of travel time which will depend upon the type of time and travel conditions.

The central VTTS parameters used in transport planning and appraisal typically refer to IVT under average or ‘normal’ conditions, and it is IVT that has received most attention in valuation exercises and also, as cited by Wardman et al. (2025), in review studies and meta-analyses. The estimates of valuations of other types of time are often expressed relative to the value of IVT, as so-called *time multipliers*. These are ratios of the marginal utility of the attribute in question and the marginal utility of IVT, and do not contain the marginal utility of money present in VTTS.

The objective of this paper is to provide the most extensive review and meta-analysis so far undertaken of a wide range of time multipliers. It builds upon previous studies (Wardman, 2004; Wardman, 2013) with much additional evidence and, for the first time, a worldwide scope.

A substantial body of evidence has long existed for time multipliers,¹ which are used in conjunction with VTTS in transport appraisal to monetise different time impacts as well as to create generalised cost metrics used in transport modelling and planning. Whilst time multipliers are therefore a fundamental feature of transport appraisal, enabling practitioners to compare and jointly consider the disparate types of time that are relevant to a policy or scheme, we are aware of only one meta-analysis of a wide range of time multipliers (Wardman, 2013). This was an extension of a meta-analysis of walk, wait and headway multipliers reported in Wardman (2001a). Other significant studies include the International Transport Forum (2014) and Commonwealth of Australia (2021) reviews of numerous multipliers in the context of public transport and the Wardman et al. (2016) review of European time multiplier evidence across modes and many attributes. With the exception of the latter three studies, reviews of time multipliers tend to focus on a single or only a few time attributes, are not always international in scope, and are often based on limited data. Examples include walk and wait time (McKnight, 1982; Waters, 1992; Wardman, 2004; Australian Transport Council, 2006; Wallis et al., 2013), travel time variability (Tseng, 2008; Li et al., 2010; Carrion and Levinson, 2012²; Wardman and Batley, 2014)³, headway (Wardman, 2001a; Wallis et al., 2013) and congested travel time (Wardman and Ibáñez, 2012).

A meta-analysis of time multipliers has several attractions. Whilst direct evidence on values of IVT that can be used for appraisal exists in many countries and contexts, or else can be deduced from relevant meta-analyses and by using conventions on how values relate to the wage rate, evidence and conventions relating to other travel attributes is less widespread. A meta-analysis of time multipliers would allow monetary valuations of these attributes to be deduced from local monetary values of IVT. Time multipliers can reasonably be expected to be more stable and more transferable across countries and contexts and over time than monetary values since variations in the marginal utility of money will cancel out when taking ratios of valuations and there is then no need to account for inflation, income growth and exchange rates. Even in appraisal contexts where evidence on valuations of time variables other than IVT is available, meta-analysis multiplier evidence could serve a useful benchmarking purpose. And although there exist multiplier conventions that can be used, a meta-analysis could query whether those which have been around for a long time, such as weighting walk and wait time at twice IVT, continue to apply. Moreover, multipliers are typically assumed to be constant, but this will not apply if either the monetary values of the non-IVT (numerator) variables or the monetary values of IVT (denominator) vary across contexts, and meta-analysis is well suited to exploring the extent to which time multipliers vary according to methodological, temporal or contextual factors which would be novel and most welcome.

Since there have been several meta-analyses of monetary valuation evidence covering both IVT and a range of non-IVT variables, implicit time multiplier equations can therefore be obtained as the ratios of the elements of the meta-analysis equation relating to the non-IVT variables relative to the element of the meta-analysis equation specific to IVT (Wardman, 2004; Abrantes and Wardman, 2011; Wardman et al. 2016). However, these equations relate to separate studies and other unaccounted for and confounding variations across studies could have a bearing, including but not limited to the adjustments needed for international data relating to exchange rates and purchasing power given that the equations relate to monetary valuations. The explicit analysis of within-study time multipliers and variations in them across studies, as reported here, provides a much more controlled context. In this respect, it is noted that the Wardman et al. (2016) study deduced time multipliers implied by the meta-analysis of a large amount of monetary valuation evidence for IVT and other time variables and found some quite large differences when comparison was made with the mean time multipliers obtained from the studies covered.

¹ For example, the convention of valuing walk and wait time at twice IVT is one of the oldest and most common of transport planning practices worldwide, stemming from the UK Department of the Environment’s pioneering Mathematical Advisory Unit Note 179 (McIntosh and Quarmby, 1970).

² This study reports a small-scale meta-analysis of reliability ratio multipliers.

³ This study reports a small-scale meta-analysis of late time multipliers.

1.2. Scope

This meta-analysis covers estimated multipliers for the following 21 attributes, with the term used in referring to them given in brackets. Attributes 1–6 are predominantly connected with public transport, 7–10 cover car transport, and 11–21 relate to delay, scheduling and reliability with a degree of emphasis on public transport.

1. Walking time, as part of accessing or egressing a main mode and rare instances where it is separately identified as part of interchanging, but also as a main mode itself (**Walk**).
2. Waiting time at the start of a journey for a main mode and rare instances where waiting time was separately identified as part of interchanging or egressing (**Wait**). In some cases, which we did not record, the multiplier would relate to waiting time specified as half the service interval, usually for shorter distance trips with higher frequency services where this is a reasonable approximation, whilst in other cases it would have been estimated to reported or specified waiting time.
3. Out-of-vehicle time (**OVT**), which is here taken to be a combination of walking and waiting time accessing or egressing a main mode and instances where the walking and waiting time elements of interchange were also included as not separately identified.
4. Access/egress time, which is the time spent getting to and from modes including any waiting time, with its unique feature being that it can cover vehicular time and is not restricted to walking time (**AccEg**).
5. Connection time, which is the difference between the arrival time of one service and the departure time of the connecting service and is made up of both the walking time and the waiting time involved in changing between public transport services, and most usually the same mode (**Conn**).
6. Headway, which is the interval between timetabled public transport services (**Headway**).⁴ Where interchange is involved, the headway is effectively the maximum of the two legs. We assume RP studies allowed for this. And we assume that in SP exercises where there is an interchange and headway is specified the respondent takes the headway to relate to the end-to-end journey opportunities which seems appropriate. Very few bus journeys involve interchange and our feeling is most SP studies that value headway for other modes are based primarily on through services.
7. Time spent searching for a parking space (**Search**).
8. Car time spent in congested traffic conditions with no distinction of the degree of congestion (**Cong-nd**).⁵
9. Car time spent in moderately congested traffic conditions (**Cong-mod**).⁵
10. Car time spent in heavily congested traffic conditions (**Cong-hvy**).⁵
11. Displacement time, which denotes departing earlier or later than the preferred departure time⁶ but with no distinction as to whether the displacement time is earlier or later (**Disp-nd**).
12. Displacement time that involves departing earlier than the preferred departure time (**Disp-earlier**).
13. Displacement time that involves departing later than the preferred departure time (**Disp-later**).
14. Mean late arrival time at the destination relative to timetable or expected arrival time that is estimated to a frequency distribution of, say, 5 or 10 late arrival times (**MeanLate-Distⁿ**).
15. Mean late arrival time at the destination relative to timetable or expected arrival time that is estimated to a given amount of late time for some proportion of journeys with the remainder being on time arrivals, such as 1 in 5 arrivals is 15 min late (**MeanLate-X/Y**).⁷
16. Early arrival at the destination relative to a preferred arrival time⁶, referred to as schedule delay early (**SDE**).
17. Late arrival at the destination relative to a preferred arrival time, referred to as schedule delay late (**SDL**).
18. The standard deviation of travel time, where the time multiplier is commonly referred to as the reliability ratio (**RR**).

According to Bates et al. (2001) and Fosgerau and Karlström (2010), under certain conditions there is an equivalence between the scheduling approach and the mean–variance (**RR**) approach. In other words, they give alternative representations of the same phenomenon. Nonetheless, some studies, including but not limited to Small et al. (1999) and Significance et al. (2012), include both terms. Whilst this is presumably because there is a feeling that there are elements of unreliability that are not captured by schedule delay terms as well as the challenges specifying the latter, we would expect lower multipliers for the separate terms when they are each entered into the model. We additionally make the following distinctions for **SDE**, **SDL** and **RR**.

⁴ Section 3.3 discusses the non-linear relationship between waiting time and headway.

⁵ Although precise definitions in terms of written descriptions and pictorial representations vary, moderate congestion is often described in terms of time slowed down by other traffic, light traffic or light congestion with heavy congestion described as stop/start traffic conditions, heavy traffic or heavy congestion. Where a single congested time variable is specified, it tends to be defined simply as congested traffic conditions. In all cases, the multiplier is expressed relative to car time spent in free flow traffic conditions.

⁶ Of course, establishing preferred departure times, which underpin 11–13, and preferred arrival times, which underpin 16–17, are not without their difficulties. Displacement time and schedule delay will be more common for timetabled transport.

⁷ When this data collection process first started, we were aware that there were few valuations relating to lateness at the departure point and hence focussed upon lateness at the destination. Whilst it turned out that evidence relating to lateness at departure became increasingly available, this was not collected since we anticipated that there would not be sufficient data. With hindsight, it would have been better to collect it. And the MeanLate-X/Y multiplier estimate might be dependent upon the stated proportion of late times but this information was not collected.

19. *SDE* when the estimated model also contained *RR* (*SDE-withRR*).
20. *SDL* when the estimated model also contained *RR* (*SDL-withRR*).
21. *RR* when the estimated model also contained *SDE* and *SDL* (*RR-withSDE/L*).

As far as we are aware, this provides the most extensive review of multipliers undertaken. Compared to the Wardman (2013) meta-analysis, this study additionally covers *OVT*, *Cong-mod*, *Cong-hvy* and *MeanLate-Dist*ⁿ. It does not cover interchange penalties or crowding valuations, both of which are important determinants of the attractiveness of public transport. Information on these variables was not collected in this or the previous reviews upon which this paper builds, largely because crowding and interchange variables were regarded to take quite diverse and less transferable forms.⁸

1.3. Structure of paper

This paper is structured as follows. Section 2 provides a summary of official guidance on time multipliers worldwide. Section 3 describes the data collection process and the key features of the data and the multipliers. The meta-analysis is the subject of section 4 whilst section 5 considers the implications of the meta-analysis by offering illustrative multipliers for a range of contexts and comparisons with official recommendations. Section 6 provides concluding remarks.

2. Summary of official time multiplier guidance

Section 1.1 pointed out that meta-analysis can serve a useful purpose in testing conventional wisdom and official appraisal guidance surrounding time multipliers. We here summarise the official multiplier recommendations provided by national or pan-national administrations, building upon those reported in Mackie et al. (2014). These are relatively easily sourced and summarised, unlike the multipliers used in modelling studies, although we note that it is not uncommon that the latter adopt the multipliers of official guidance. In other words, official guidance not only steers appraisal practice but also modelling practice. We refrain here and elsewhere from consideration of and comparison with the results of specific studies since it is the purpose of this study to review the relevant empirical evidence in its entirety.

Table 1 presents the official multiplier recommendations of which we are aware. Only Australia provides recommended displacement time multipliers, of 0.33 for earlier and 0.50 for later, whilst it is noted that some countries use fixed penalties where interchange occurs instead of connection time multipliers. The countries that provide recommendations tend to be those that undertake so-called national value of time studies to underpin that guidance and which, as is subsequently apparent in Table 4, provide most of the valuation evidence reviewed here.

There are 12 instances of *Walk* multiplier recommendations. The long-established convention of weighting *Walk* at twice *IVT* features for four administrations, with two each of 1.5 to 2 and of 1.5. The others tend to be in the range 1 to 1.5. Some administrations provide a *Wait* recommendation or a *Headway* recommendation but not both, on the grounds that different service headways have impacts in terms of waiting time. There are 12 *Wait* multiplier recommendations with five being 2, two in the range 1.5 to 2 and three of 1.5, with the remainder mainly between 1.1 and 1.4. There are recommendations for *Conn* multipliers in 6 cases and they centre around 1.5, although presumably *Walk* and/or *Wait* multipliers are used where there are no specific *Conn* multiplier recommendations. It seems that the customary weight of 2 for *Walk* and *Wait* is not quite as dominant as might be envisaged.

The recommended *Headway* multipliers tend to depend upon service interval, and to fall as services become less frequent. They tend to vary across a large range, being as high as 1 and as low as 0.2. However, these multipliers are not purely service headway effects, and at higher frequencies will represent waiting time more.

A development in recent years, presumably due to methodological advances and perhaps an increasingly unreliable world, is the amount of evidence relating to travel time variability, and 10 of the 15 administrations provide recommendations. There is a tendency for the *RR* to be used for unscheduled modes and late time (*LT*) for scheduled modes. The *LT* values at the destination tend to lie in the range 2.5 to 3.5 but, surprisingly, with larger valuations when there is lateness at the departure. The *RR* multiplier exhibits a large range from around 0.1 through to 1.4. Notably there is much more segmentation of the multipliers relating to travel time variability than the other multipliers.

Finally, a few administrations recommend congestion time multipliers. Two make no distinction between the degree of congestion and both recommend 1.5. The Norwegian values which distinguish moderate and heavy congestion tend to vary around 1.5 but the multipliers for light and heavy congestion in Singapore are both below this.

⁸ For example, crowding has sometimes been represented in qualitative terms whilst the pure interchange penalty will depend upon whether it is the sole representation of interchange or specified alongside connection time. The valuations of both attributes are very context specific, influenced strongly by the design of the vehicle/facility in which crowding/interchange occurs and the consequences of missing a connection. The amount of usable evidence for each attribute is relatively limited.

Table 1

Summary of official multiplier guidance.

	Walk (AccEg)	Wait	Conn	Headway	Mean Late Time (LT) ^a or RR	Congestion
Australia	1.5	1.4	1.5	0.93 at 5 min 0.37 at 60 min	LT: 5.9 PT departure LT: 2.8 PT LT: 4.1 PT average	
Chile	2.0	2.0				
Denmark		2.0	1.5	0.8	LT: 3.0 PT LT: 1.5 Car and Cycle	
EU	1.5 – 2.0	1.5 – 2.0		0.4 – 1.0	LT 2.5 – 4.0 PT RR 0.4 – 1.2 Car	1.5
France	2.0	1.5	2.0		Bespoke approach ^b	
Germany		Bespoke approach ^c	1.3		LT: 2.7	
⁵ Ireland ^d	1.5 – 2.0	1.5 – 2.0			LT: 2.9 PT Commute LT: 2.8 PT Business LT: 2.5 Other Bus/LRT LT: 3.2 Other Rail RR: 0.8 Car	
Netherlands	1.3 Business 1.3 Shopping 1.8 Commute 1.9 Other	1.5			RR: 0.21 – 0.35 Car RR 0.11 – 0.32 Rail RR: 0.56 – 0.65 Local PT RR: 0.28 – 0.30 Air (vary by purpose)	
New Zealand		2.0		1.00 at 5 min 0.40 at 60 min	Depart: Arrival LT: All 5.0:2.8 LT: Train 3.9:2.4 LT: Bus 6.4:3.2 LT: Commute 5.5:2.8 LT: Education 3.0:3.8 LT: Other: 5.4:2.0	

(continued on next page)

Table 1 (continued)

	Walk (AccEg)	Wait	Conn	Headway	Mean Late Time (LT) ^a or RR	Congestion
Norway	1.3	Based on headway	1.2	1.07 at 0–15 min 0.98 at 16–30 0.63 at 31–60 0.47 at 61–120 0.18 at > 120		Car drivers: Moderate: Heavy ^e Business: 1.1:1.4 Commute: 1.2:2.3 Leisure: 1.3:2.4
Singapore	1.3–1.5 Bus 1.1–1.3 Metro 1.0 Taxi	1.2–1.4 Bus <1 Metro 1.1 Taxi				Light 1.06 Heavy 1.35
Sweden	0.8 (<30 min) 1.4 (>30 min)			Short Distance: 1.15 at <10 m 0.14 >120 m Long Distance: 0.52 to 0.20 EB: 0.7 to 0.42	LT: 3.5 Scheduled RR: 0.9 Unscheduled	1.5
United Kingdom	2.0	2.0	1.5 Rail	0.64	LT: 2.5 Rail RR: 1.4 Car	
United States	2.0	2.0				
World Bank	1.5	1.5				

Notes: ^a Lateness at destination unless otherwise stated. ^b [Ministère de la Transition écologique, de l'Énergie, du Climat et de la Prévention des Risques \(2019\)](#). ^c [Intraplan and VWI Stuttgart \(2022\)](#). ^d *Walk*, *Wait* and *Conn* multipliers do not feature in the main guidance document but are used in their 'Regional Modelling System' based on UK evidence. ^e These are relative to somewhat average traffic conditions. The guidance also includes multipliers for 'free flow' conditions, ranging between 0.8 and 1.

3. The Data

3.1. Data collection process

The multiplier evidence is drawn from studies that involve the modelling of travellers' choices, covering Revealed Preference (RP) data based upon actual choice contexts or various forms of Stated Preference (SP) dealing with hypothetical choices.⁹

This study builds upon the meta-analysis reported in Wardman (2013) that examined 1389 time multipliers from 244 European studies and 18 countries covering the years between 1964 and 2011¹⁰ and which were drawn from a dataset primarily collected to undertake meta-analysis of monetary valuations (Wardman et al., 2016). The additional multiplier data used in this study were collected as part of a World Bank funded study (Wardman et al., 2025) with an original aspect, compared to previous reviews, of extension to worldwide evidence.

Compared to Wardman (2013), there is here much more data. It is drawn from 606 (+148%) studies worldwide that yield 2872 (+107%) time multipliers covering 52 (+189%) countries with data collected between 1963 and 2023, and with coverage extending to additional attributes as set out in section 1.2. The full list of studies covered is provided in a supplementary file. As for the potential explanatory variables, the following information was collected for each multiplier observation, with further detail reported in Table 6.

- year of data collection and country;
- the number of observations in the estimated model¹¹;
- whether the trips were urban, inter-urban or a mix of the two, along with an indicator of urban hierarchy;
- journey purpose;
- the modes used by the sample of travellers analysed and the modes to which the estimated values related;
- the type of data to which the valuation was estimated, which is either RP, SP or a combination of the two;
- the choice context examined and estimation method employed;
- where an SP exercise was used, the number of alternatives, attributes and choice situations offered, along with the means of presentation, experimental design procedure and means of customisation;
- the purpose of the study and whether the main aim was the valuation of the variable in question;
- the source of the evidence, covering academic journals, conference papers, published reports, and unpublished reports of various types;
- survey method, including on-mode, roadside or on-street interviews, interviews at stations/airports, hall tests, telephone contacts, home or workplace interviews, managed online panel, intercept, unsolicited mail-out, and use of mailing lists/social media; and
- whether the multiplier was obtained from a study that included a cost term or just trade-offs between time-related variables.

A variety of means were used to identify the additional studies. These included: the use of search engines, such as Google Scholar, Scopus and Web of Science; the systematic search of numerous academic journals and accessible conference proceedings; and contacting academics and consultants active in the field. The reference lists of included documents were examined (backward snowballing) and documents that subsequently cited identified research were followed up (forward snowballing).

The dataset was reviewed to identify and remove duplicate versions of the same study prior to proceeding to the main analysis. Results obtained from re-analysis of the same dataset were permissible if they provided new insights.

3.2. Key features of the assembled data

The source of the evidence is summarised in Table 2. Peer-reviewed journal articles are the most common source, and along with conference papers (some of which may have been published subsequently in journals) and published reports form the majority of the evidence. Nevertheless, given that unpublished reports ('grey literature') commissioned by governmental organisations, in the UK in particular, provided important and extensive evidence for our previous meta-analysis datasets, which have been used as a starting point for the present research, these still form a significant proportion of the values and studies, though in some cases they too will have been published subsequently.

Multiple observations were extracted per study either because the values relate to different attributes or else they are separate values for the same attribute but segmented by key influential variables such as distance, mode, journey purpose or data type and estimating the impacts of these variables is central to our investigation. Table 3 provides a summary. The average number of valuations

⁹ Multiplier estimates in the dataset do not include official or expert recommendations. We cannot assess, however, whether some of the estimates were influenced by expert judgement during the studies themselves, although we feel that such judgement is more likely to impact VTTS than time multipliers.

¹⁰ A prior meta-analysis of 469 *Walk*, *Wait* and *Headway* multipliers covering British evidence between 1964 and 2000 was reported in Wardman (2001a).

¹¹ The primary motivation for assembling the data over several different review studies was to investigate **monetary** valuations. Confidence intervals for monetary valuations were not collected but only 'retro-fitted' as part of a recent study (Wardman et al., 2025). Confidence intervals for time multipliers were not retro-fitted and although some could be deduced approximately from those for the monetary values they would be far from comprehensive.

per study is 4.7, with 20% of studies yielding only a single valuation and a majority (55%) providing three or less. Only 10% of studies delivered more than 10 multipliers, with 7 (1%) providing more than 25 and the maximum being 50.

The numbers of valuations and studies for each country are listed in Table 4, along with the World Bank income categorisation of high (H), upper-middle (UM) and lower-middle (LM). The ordering is according to the number of multiplier observations.

Given this study builds upon previous UK meta-analyses of monetary valuations of time-related attributes and a subsequent meta-analysis of European evidence, it is hardly surprising that the UK has the largest proportion of valuations (36%) and of studies (30%).¹²

The economic appraisal of transport schemes is common in Western European countries, as is their ongoing commitment to national studies, and Denmark, Netherlands, Norway, Sweden and Switzerland provide 23% of valuations and 18% of studies. Australia, with its rich heritage of transport research, provides 9% of valuations and 9% of the studies whilst Singapore has conducted several national studies which have a strong public transport focus and yield a relatively large number of multipliers per study.

The perhaps surprisingly low number of valuations for the USA and the relatively few valuations per study for both the USA and Canada presumably stem from the dominance of the private car where the multipliers of interest here tend to be less relevant.

The UM countries provide 5% of valuations and 9% of studies whilst the respective figures are lower at 2% and 4% for the LM countries. The examination of valuations for attributes other than IVT would seem, up to now, to be a luxury of developed countries.

RP data provided 12% of the values, with 5% from joint RP-SP models and the remaining 83% from some form of SP exercise. 3% and 1% of the RP evidence is from UM and LM countries respectively, not greatly different to the 5% and 2% for the non-RP evidence. All countries provided non-RP evidence, three quarters of high income countries provided RP evidence but only a third of UM and LM countries did so. As to whether the estimation of the valuation in question was deemed to be the main purpose of the study, as opposed to incidental, these form around a third of all valuations.

Multipliers estimated to data collected up to 1980 provide only 2% of the valuations, with the years 1981–1990, 1991–2000, 2001–2010, and 2011 onwards respectively accounting for 11%, 27%, 35% and 25% of valuations. The corresponding pattern for the number of studies is little different at 4%, 11%, 25%, 32% and 28% respectively. Some increase over time would be anticipated but it might be argued that an element of saturation, more likely in developed countries, has been achieved on the grounds that the cumulative knowledge base, more widespread official recommendations and the increased number of review studies reduce the need for further studies.

3.3. Summary multiplier valuations

We now turn to the valuations themselves. Table 5 reports descriptive statistics for the 21 attributes set out in section 1.2. It contains the mean value for each attribute, the standard error of the overall mean (SE) and associated 95% confidence interval (CI), the sample size (Obs) and number of studies from which the multipliers are drawn, the median and the sample skewness.¹³ The mean (Mean13), standard error (SE13) and sample size (Obs13) for the Wardman (2013) study upon which this study builds are provided in the final three columns. It is the purpose of the meta-analysis to explain variations in multipliers and hence no segmentations are here provided.

Starting with the sample sizes, *Walk*, *Wait* and *Headway* form by far the largest number of observations, consisting of 19%, 14% and 18% respectively. *AccEg* and the various forms of congested time contribute a further 10% and 12% respectively. There are relatively few observations for *MeanLate-Distⁿ* (0.5%), *Search* (1.4%), and for *SDE-withRR*, *SDL-withRR* and *RR-withSDE/L* combined (1.5%).

The valuations have a positive skew for all but the smallest category of *SDE-withRR*. The mean is greater than the median in all these cases except for *Search*. However, the degree of skewness is generally not great, being largest for *AccEg* and the various congestion multipliers and reflecting a few very large valuations, in excess of 3, for these variables.

The mean values for the multipliers for *Walk*, *Wait*, *AccEg* and *Conn* are remarkably similar, ranging between 1.65 and 1.72, with a slightly higher mean of 1.87 for *OVT*. The median values are lower, between 1.50 and 1.55, with 14% of the multipliers less than one and only 3% less than 0.8. These overall results contrast with the convention of valuing *Walk* and *Wait* at twice *IVT*.

The *Headway* multiplier averages 0.60 which seems credible. To interpret this figure in more detail, relationships can be formulated between *Headway* and *Wait* for high frequency services and between *Headway* and displacement time for less frequent services. For high frequency services, travellers can be expected to randomly arrive for their departure and then, given that on average waiting time would be half the service headway, the *Headway* multiplier would be half of the *Wait* multiplier. At 5 min service frequencies it can be assumed that everyone would be a random arriver and hence, given a *Wait* multiplier of 1.72, the implied *Headway* multiplier would be 0.86. This serves as an upper bound, and the mean *Headway* multiplier in Table 5 is consistent with it. As services become less frequent, the link between waiting time and headway is weakened as it becomes more likely that travellers plan their journeys according to scheduled departure times and simply arrive a number of minutes earlier, depending on various factors such as their risk aversion and the consequences of missing the service. The inconvenience of service headway is then represented by this 'planning penalty' plus the

¹² And it should be recognised that the UK has a long history of using valuations of a wide range of travel attributes in transport planning and appraisal.

¹³ The sample skewness is calculated as $[N \bullet \sum (X_i - \bar{X})^3] / [S^3 \bullet (N - 1) \bullet (N - 2)]$, where $\bar{X}$ is the sample mean, S is the sample standard deviation and N is the sample size.

Table 2
Source of Evidence.

Source	Values	Studies
Journal	879 (31%)	264 (44%)
Conference	482 (17%)	105 (17%)
Published Report	328 (11%)	32 (5%)
Unpublished Operator Commissioned	250 (9%)	46 (8%)
Unpublished Government Commissioned	796 (28%)	125 (21%)
Unpublished Academic Report	130 (4%)	32 (5%)
Unpublished In-house Report	7 (0%)	2 (0%)
Total	2872 (100%)	606 (100%)

Table 3
Valuations per study.

Values	Studies (%)	Values	Studies (%)
1	123 (20%)	7–8	43 (7%)
2	154 (25%)	9–10	24 (4%)
3	60 (10%)	11–15	38 (6%)
4	77 (13%)	16–25	16 (3%)
5–6	64 (11%)	>25	7 (1%)

Table 4
Values and studies per country.

Country	Values	Studies	Country	Values	Studies
UK (H)	1041	184	Mexico (UM)	14	4
Australia (H)	258	52	Portugal (H)	13	4
Netherlands (H)	248	44	Finland (H)	11	3
Denmark (H)	125	13	Saudi Arabia (H)	9	2
USA (H)	119	55	Taiwan (H)	8	4
Norway (H)	109	17	Greece (H)	8	3
Singapore (H)	100	6	Bangladesh (LM)	7	4
Sweden (H)	88	19	Czech Republic (H)	7	2
Switzerland (H)	86	14	Thailand (UM)	5	4
New Zealand (H)	81	10	Serbia (UM)	5	1
Spain (H)	60	16	Ireland (H)	4	3
Chile (H)	50	18	Panama (UM)	4	2
China (UM)	49	19	Brazil (UM)	4	1
Colombia (UM)	43	15	Latvia (H)	4	1
South Korea (H)	37	7	UAE (H)	4	1
France (H)	37	6	Pakistan (LM)	3	2
Canada (H)	32	15	Vietnam (LM)	3	2
Germany (H)	29	8	Hong Kong (H)	3	1
Italy (H)	28	11	Peru (UM)	3	1
India (LM)	19	7	Bulgaria (UM)	2	1
Japan (H)	19	7	Lebanon (UM)	2	1
South Africa (UM)	19	7	Myanmar (LM)	2	1
Austria (H)	17	3	Russia (UM)	2	1
Belgium (H)	16	6	Argentina (UM)	1	1
Indonesia (LM)	16	6	Ghana (LM)	1	1
Israel (H)	16	5	Nigeria (LM)	1	1

Note: The number of studies here sum to 622 whereas the total number of studies in our dataset is 606. This is because 7 studies covered 2 countries, 2 studies covered 3 countries and 1 study covered 6 countries.

displacement time incurred.¹⁴ Assuming a uniform distribution of preferred departure times, the displacement time would be a quarter of the service headway. At a service headway of hourly the vast majority would be expected to plan their departure. If travellers arrive 5 min before departure and with a mean *Wait* value of 1.72, and given *Disp-earlier*, *Disp-later* and *Disp-nd* of around 0.6, a *Headway* multiplier of 0.29 is implied. This increases to 0.44 with arrival 10 min before departure. These figures serve as a lower bound.

We might reasonably expect the headway multiplier to be non-linear with respect to service frequency, but we are not in a position to examine this since the requisite data was not collected. But overall, our evidence on the *Headway* multiplier is consistent with the

¹⁴ Fosgerau (2009) establishes a relationship between headway valuations and displacement time but it is not readily operationalised with the summary evidence of Table 5.

Table 5
Descriptive statistics for multiplier valuations.

Variable	Mean	SE	95% CI	Obs	Studies	Median	Skew	Mean13	SE13	Obs13
<i>Walk</i>	1.67	0.030	1.61–1.73	549	215	1.55	0.91	1.68	0.04	344
<i>Wait</i>	1.72	0.038	1.64–1.80	408	185	1.53	0.96	1.80	0.07	138
<i>AccEg</i>	1.70	0.051	1.60–1.80	292	110	1.50	2.27	1.68	0.07	144
<i>OVT</i>	1.87	0.104	1.66–2.08	99	45	1.54	0.84	–	–	–
<i>Conn</i>	1.65	0.072	1.51–1.79	75	30	1.53	0.89	1.84	0.10	26
<i>Headway</i>	0.60	0.014	0.57–0.63	514	146	0.57	0.83	0.71	0.03	329
<i>Search</i>	1.60	0.097	1.41–1.79	40	19	1.62	0.87	1.85	0.21	24
<i>Cong-nd</i>	1.48	0.036	1.41–1.55	192	55	1.35	1.96	1.56	0.06	75
<i>Cong-mod</i>	1.31	0.054	1.20–1.42	69	19	1.19	3.03	–	–	–
<i>Cong-hvy</i>	2.09	0.133	1.82–2.36	69	19	1.73	2.59	–	–	–
<i>Disp-earlier</i>	0.54	0.030	0.48–0.60	97	38	0.49	0.71	0.56	0.07	44
<i>Disp-later</i>	0.68	0.037	0.61–0.75	101	38	0.56	0.76	0.65	0.08	47
<i>Disp-nd</i>	0.58	0.067	0.45–0.71	18	7	0.44	0.41	0.74	0.14	16
<i>MeanLate-Distⁿ</i>	2.53	0.133	2.26–2.80	15	2	2.51	0.21	–	–	–
<i>MeanLate-X/Y</i>	3.12	0.233	2.65–3.59	66	27	2.47	1.34	4.10	0.44	37
<i>RR-withoutSDE/L</i>	0.93	0.065	0.80–1.06	149	39	0.61	1.65	0.91	0.10	45
<i>RR-withSDE/L</i>	0.71	0.135	0.44–0.98	15	4	0.53	1.08	–	–	–
<i>SDE-withoutRR</i>	0.96	0.101	0.76–1.16	34	16	0.86	0.83	0.81	0.07	54
<i>SDL-withoutRR</i>	2.23	0.147	1.94–2.52	42	22	2.20	1.18	1.70	0.11	66
<i>SDE-withRR</i>	0.31	0.053	0.20–0.42	7	3	0.31	–0.93	–	–	–
<i>SDL-withRR</i>	2.00	0.251	1.50–2.50	21	5	1.93	1.27	–	–	–

evidence on *Wait* and *Disp-earlier*, *Disp-later* and *Disp-nd*.

Search has a premium valuation for car users a little lower than the *Walk* and *Wait* multipliers which apply mainly to public transport use, and this seems reasonable to reflect the annoyance involved in searching for a space to park. Congested time will have a multiplier in excess of one due to the frustration, greater difficulty and heightened risk involved in driving in such conditions relative to free flow. The results are as anticipated in implying that *Cong-hvy* incurs a higher disutility than *Cong-mod*, with *Cong-nd* between the two. And only 5% of the congestion multipliers were less than 1 with a minimum of 0.77.

The late time multipliers indicate the adverse consequences of arriving late and, as might reasonably be conjectured, are on average larger than the disutilities associated with *Walk* and *Wait*. Only one late time multiplier was less than 1. The multiplier for *MeanLate-X/Y* is noticeably larger than that for *MeanLate-Distⁿ*, and a possible explanation here is that not all studies were explicit in stating that on Y-X occasions the arrivals would be on time. If respondents felt that there would still be an element of lateness associated with the Y-X occasions, this would serve to inflate the estimated lateness multiplier. On the other hand, the median values are very close for *MeanLate-X/Y* and *MeanLate-Distⁿ*, at around 2.5.

What [Table 5](#) terms *SDL-withoutRR*, *SDE-withoutRR* and *RR-withoutSDE/L* denote multipliers that relate to SDL, SDE and RR when estimated without the additional inclusion of RR and SDE/L. *SDL-withoutRR* denotes lateness relative to the preferred arrival time and has amongst the highest multipliers, reflecting the strong dislike of lateness, but it is lower than for *MeanLate-X/Y* and *MeanLate-Distⁿ*. There might be confounding effects at work here, such as *SDL* being less likely to apply to public transport, being more difficult to convey or being a comparatively recent term. As seems likely, *SDE-withoutRR* has a lower multiplier, and it is insignificantly different to 1.

The *RR-withoutSDE/L* multiplier is less easy to interpret. A useful relationship is that set out by [Bates et al. \(2001\)](#). Assuming departure times to be continuously variable, as with car travel, the *RR* can be related to the values of mean time (α), *SDE* (β) and *SDL* (γ) as follows:

$$RR = \frac{\beta}{\alpha} \ln \left(1 + \frac{\gamma}{\beta} \right) \quad (1)$$

Using the mean figures from [Table 5](#) of 2.23 for γ and 0.96 for β , with α implicitly 1, yields an *RR* of 1.15 which corresponds reasonably well with the mean *RR-withoutSDE/L* of 0.93. Note though that over 10% of the *RR* observations exceeded 2 and 50% were less than 0.61 and these could not be supported by this equation with sensible *SDE* and *SDL* multipliers. Unsurprisingly, the multipliers for *RR-withSDE/L*, *SDE-withRR* and *SDL-withRR* are somewhat less than those for *RR-withoutSDE/L*, *SDE-withoutRR* and *SDL-withoutRR*.

The multipliers relating to travel time variability tend to have a large spread. On the one hand, this might be because this emotive issue attracts a disproportionate amount of protest response, whilst on the other hand, the issues being dealt with are more complex which may have a bearing on the extent to which respondents understand the SP exercise and fully appreciate the late time consequences.

Displacement time might in some circumstances be relatively highly valued, such as having to get up earlier in the morning for the journey to work or a later departure that would imply difficulties in getting to work on time. In the reverse direction, a fixed work end time would mean earlier departures are not possible and later departures would not be an attractive option, and hence each would imply relatively large valuations. But in the absence of such constraints, displacement time might be regarded to be a relatively minor inconvenience and certainly valued less than *IVT* since departing X minutes later but with an X minute quicker journey time and hence the same arrival time would be expected to be preferred to the prevailing situation. The general expectation is that displacement time is

valued somewhat less than IVT and this is the case in the data collected, with average multipliers for *Disp-earlier* and *Disp-later* of 0.54 and 0.68 respectively and between the two for *Disp-nd*. The vast majority of the multipliers are less than 1.

As for comparison with the average values reported in Wardman (2013), the differences tend to be minor, with on average the new figures 0.08 smaller and only *Headway* ($t = 3.3$) and *SDL-withoutRR* ($t = 2.9$) significantly different at the 5% level. The largest proportionate increase and reduction compared to the previous study are 31.2% and -23.9% for *SDL-withoutRR* and *MeanLate-X/Y*.

Whilst the vast majority of the multiplier evidence lies within a plausible range, there are inevitably some valuations that are somewhat outside what might be deemed plausible, most likely reflecting unreliable evidence and arising because the data collection focussed on monetary values with little attention paid to the credibility of the time multipliers. Given that the removal of ‘outlier’ observations can be contentious, the sensitivity of the mean values to different data exclusion criteria was investigated. We removed observations that were outside ± 2.58 and ± 1.96 standard deviations of the mean, representing 1% and 5% of observations, as well as observations deemed unreasonable compared to the conventional wisdom regarding the variable in question. Reassuringly, the mean values were not particularly sensitive to the removal of observations according to these criteria.

Although there are inevitably specific observations in our dataset which do not accord with theory, convention or reasonable expectation, in general the data assembled yields sensible mean values which are not very sensitive to the omission of what might be deemed outlier observations. Most multipliers lie within a relatively narrow and acceptable range. Hence we see no reason to omit data at the outset from the meta-analysis to which we now turn.

4. Meta-analysis

Meta-analysis is a specific form of empirical evidence review, here regression based, that seeks to explain variations across studies in parameters of interest, here travel time multipliers, in terms of a range of relevant explanatory factors. This multivariate approach has advantages over conventional literature reviews since it is more resilient to the confounding factors that can afflict simpler approaches, such as cross-tabulation. And the quantification of the impacts of influential variables provides methodological and policy-oriented insights along with transferable context-specific estimates of multipliers which are useful for benchmarking alongside existing evidence and even more valuable where none exists.

The reported meta-models are multiplicative, since a MacKinnon, White, and Davidson (MWD) test (MacKinnon et al., 1983) found them to be statistically superior to an additive function whilst yielding generally more precise coefficient estimates. As such, the multiplier dependent variable is specified in logarithmic form.

We could carry out the regression analysis separately for each attribute. However, the approach adopted here is to pool data across multipliers in a single simultaneous estimation since this is more parsimonious than estimating separate models, given that some impacts will be common across multipliers, and to be able to account for study-level fixed-effects as many studies provide more than one multiplier. Our pooled model is specified as:

$$\ln(M_{ijs}) = \tau_J + \sum_{j'=1}^{J-1} \alpha_{j'} \cdot \delta_{j,j'} + \sum_{j'=1}^J \sum_{k=1}^K \sum_{l=1}^{L_K-1} \beta_{j'l}^k \cdot \delta_{l,l}^k \cdot \delta_{j,j'} + \sum_{s=1}^{S-1} \gamma_s \cdot \delta_{s,s'} + \mu_{ijs} \quad (2)$$

The model explains variations in multipliers (M) relating to observation i , attribute j and study s . To allow for the simultaneous estimation over all attributes j , $J = \text{Walk}$ is arbitrarily selected as the base attribute, represented by the constant term τ_J . The term $\delta_{j,j'}$ is the standard Kronecker delta, having the value 1 if $j = j'$, 0 otherwise.¹⁵ Parameters $\alpha_{j'}$ denote the proportionate effect of a particular attribute j' relative to the base attribute J . We report these, alongside τ_J , as *main effects*.

In addition, a range of k factors which might influence the multipliers are considered via what we term *segmentation* effects (k factors that can only be drawn from the list of available explanatory variables detailed in 3.1). For each factor k (e.g. “data type”), one level l (e.g. “data type = SP”) is arbitrarily selected as the base level (denoted as L_K). An analogous Kronecker delta formulation ($\delta_{l,l}^k$) represents the inclusion of the segmentation effects, where $\beta_{j'l}^k$ denote the proportionate effect on a multiplier j of some level l of factor k , relative to the factor base level chosen (L_K).

Study-specific fixed-effects are captured by γ_s , included in response to the suggestion of one of the reviewers. These might be useful in controlling for the impact of poor quality studies, discerning unidentified factors specific to a study over and above the random error term,¹⁶ and capturing any systematic differences across the multipliers of a given study. They are ‘catch-all’ terms covering all multipliers within a specific study. Finally, the error term μ_{ijs} represents the net effect of unknown or unmeasurable influences on a multiplier observation in the estimated models.

A thorough and systematic process was followed to identify the impact of the numerous candidate segmentation variables on the multiplier valuations and to determine which of the $\beta_{j'l}^k$ in Eq. (2) should be retained. Table 6 lists the k segmentation variables investigated along with their l levels and the number of observations in each category. Initially, each segmentation variable was examined in isolation; so, for example, we examined the impact of the journey purpose categories on each multiplier and took a view as to whether the estimated effects were considered to show potential to be taken forward into pooled segmentation models. The criteria

¹⁵ There are no continuous variables in the reported *meta*-models.

¹⁶ This might include publication bias which represents a tendency for studies to report models or to be accepted for publication where the key parameters accord with the ‘conventional wisdom’.

for determining potential for further investigation was that at least one effect with significance at 10% or better could be recovered and that the variation had a possible explanation.

Clearly, when the various separate segmentation effects deemed to merit further consideration are included simultaneously within a single model the precision with which they are estimated tends to fall. Starting with the model containing all the potential effects, we first removed those where the *t* ratio was less than 1 and re-estimated the model, which will tend to increase the precision of the remaining coefficient estimates. We erred on the side of caution and retained some where an effect was expected in the hope that subsequent models would find them more significant. The process was iteratively repeated, excluding coefficient estimates with *t* ratios less than 1.40, 1.64 and finally 1.96, but again exercising a degree of judgement on retaining some coefficients. In addition, some selective tests of hypotheses were undertaken as follows.

Firstly, we tested if the *Walk* multiplier varied according to whether the walking time was a main mode or was part of using other modes. There was very little variation according to this dimension.

Secondly, we tested whether the *Walk* multiplier was impacted by whether a country could ‘undisputably’ be categorised as being high temperature, on the grounds that then walking is more arduous.¹⁷ The incremental effect was not remotely significant.

Thirdly, a pattern that is sometimes found in empirical studies (Li et al., 2010; Hensher, 2015)¹⁸ is that the VTTS is lower if reliability is controlled for, the intuition being that without such control the VTTS also includes some unreliability effect. This would imply larger time multipliers. There are 175 observations of non-congestion multipliers in the data set where reliability was also included in the estimated model. The incremental effect for this segmentation was far from significant. We also split out 49 observations of congested multipliers since we might expect these to be lower where reliability is included, on the grounds they would otherwise discern a reliability effect. We found a significant effect with the congestion multipliers 22% larger when the reliability terms were also included but this is contrary to expectation and hence not retained.

Finally, we examined whether multipliers were different if they were obtained from studies which included a cost variable or where the trade-offs were solely between different types of time. This tests the hypothesis that biases in SP responses, particularly strategic bias, are most likely to focus on the cost term and therefore the impacts cancel out when estimating time multipliers. When the SP exercise does not contain cost, it is possible that the attention switches to travel time, which will then impact the time multipliers. No significant differences were detected.

As part of this investigation of segmentation effects, we also specified some composite terms, on the grounds of expected or actual similarity of effects or because of small sample sizes. These composite terms were:

- *WWAOC* which is *Walk*, *Wait*, *AccEg*, *OVT* and *Conn* combined;
- *Late* as an amalgamation of the three related variables of *MeanLate-Distⁿ*, *MeanLate-X/Y* and *SDL*;
- *Rely* which is a combination of *Late* and *RR*;
- *CongSearch* which combines the car specific terms of *Cong-nd*, *Cong-mod*, *Cong-hvy* and *Search*;
- *Disp* which covers *Disp-earlier*, *Disp-later* and *Disp-nd*;
- *WAC* which is *Walk*, *AccEg* and *Conn* combined.

Table 7 reports the results of our preferred selection of meta-models. The coefficient estimates along with *t* ratios in parentheses (Coeff (*t*)) are reported as well as the percentage variation (%Effect). For the main effects, %Effect is specified relative to the base category of *Walk*, represented by the constant τ_j , and we provide for reference the implied multiplier for *Walk* after the adjustment of the constant, discussed in section 5, by adding half of the model’s mean square error (MSE) to it. For the segmentation effects, %Effect is relative to all non-included categories for the main effect multiplier being segmented. The notation for the segmentation effects specifies the main effect (*j*) that is segmented followed by the level (*l*) of variable *k* that forms the segmentation and is further described when we come to discuss the results (e.g., *Wait_{RP}* gives the incremental effect of RP data type on the *Wait* multiplier).

Model I, which omits the fixed-effects, contains those coefficient estimates that were significant and correct sign at the usual 5% significance level as well as some which were not significant but deemed to merit retention. This model achieves an $\bar{R}^2$ goodness of fit only slightly larger than the 0.507 when no segmentation effects were specified. The correlations between estimated coefficients are encouragingly low; the only ones to exceed 0.5 in absolute terms were between *Cong-nd* and *CongSearch_{NonMetropolis}* (−0.63), *Disp-earlier* and *Disp_{UseCar}* (−0.56) and *Disp-later* and *Disp_{UseCar}* (−0.54). Looking beyond pairwise correlations, tolerance parameters determine if multicollinearity across the full set of included variables is an issue. This involves regression of each independent variable on all other independent variables and $1-R^2$ is each variable’s tolerance level. A tolerance level less than 0.2 is considered to indicate strong multicollinearity (O’Brien, 2007) and the lowest here were 0.19 for *RR*, 0.33 for *Headway* and 0.33 for *CongSearch_{NonMetropolis}*, with the remainder greater than 0.4 and 75% greater than 0.65. The extent of multicollinearity is therefore very limited.

As for the main effects in Model I and indeed the other two reported models, the parameters relating to *Wait*, *AccEg*, *OVT* and *Conn* were all far from significantly different relative to the base of *Walk*, with *Search* a little lower. Although the overall implied multipliers for these variables will depend upon the segmentation effects, when we estimated simple models without any segmentation terms we

¹⁷ We did not have resource to do anything more sophisticated, but then even this clear dichotomy did not reveal anything.

¹⁸ However, the most recent UK national study (ARUP et al., 2015) found larger VTTS when reliability terms were included in the estimated SP models.

Table 6
Segmentation variables considered, categories and sample sizes.

Variable	Categories and Multiplier Observations	Selected for Further Analysis?
Data Type	RP (342), SP (2397), Joint RP-SP (133)	Yes
Mode Used	Car (970), Bus (347), Train (332), LRT/Metro (98), Taxi (25), Air (31), Various PT (222), Car and PT (652), Others and Combinations (195)	Yes
Mode Valued	Car (737), Bus (411), Train (438), LRT/Metro (146), Taxi (25), Air (38), Various PT (541), Car and PT (344), Others and Combinations (192)	No
Year	Up to 1990 (363), 1991–2000 (775), 2001–2010 (1016), 2011 onwards (718)	Yes
Spatial Dimension	Urban (2064), Inter-Urban (527), Both Urban and Inter-Urban (281)	Yes
Urban Hierarchy	Major Metropolis (534), Metropolitan (558), Urban (785), Rural (139), Mix (48), Not Relevant (808)	Yes
Journey Purpose	Business (242), Commuting (1005), Leisure (708), Non-Business (426), Non-Commuting (79), All (404), Other (8)	Yes
Sample Size	Number of observations in estimated model, and ≤1000 (752), 1001–2500 (726), 2501–5000 (639), >5000 (755)	No
Choice Context	Mode Specific (1023), Mode Choice (1461), Route Choice (194), Mode and Destination Choice (30), Time of Day Choice (28), Station Choice (17), Speed/Lane Choice (29), Combinations (90)	Yes
Type of Study	Valuation (1582), Forecasting (1138), Methodological (152)	Yes
SP Complexity	Number of Variables, Alternatives, Scenarios Offered	No
SP Presentation	Pen and paper (848), Cards/Show Material (209), Computer Assisted (787), Computer Web (584), Phone (48), Face-to-face (17), Smart phone (13), Not Specified (24), Not Applicable – RP (342)	Yes
SP Customisation	Hypothetical journey (82), Tailored Absolute Design (894), Pivot using Computer (878), Pivot using ‘Changes from Now’ (161), Homogeneous Trips (144), None (90), Not Specified (281), Not Applicable – RP (342)	No
SP Design Procedure	Orthogonal (1495), Efficient (477), Boundary values (227), Bradley design (61), Adaptive (36), ‘Ad hoc’ (21), Random (34), Not Specified (179), Not Applicable – RP (342)	No
Model Type	Multinomial Logit (2075), Nested Logit (230), Mixed Logit (467), Hybrid (61), Other (39)	No
Taste Variation	Systematic Variation (182), Normal Distribution (115), Lognormal Distribution (166), Triangular Distribution (77), Other Distributions (49), Systematic and Random Taste Variation (42), Latent Class (55), None (2186)	Yes
Survey Method	On-mode (260), Roadside or On-street Interviews (42), Station/Airport/Shopping Centre/Car Park Interviews (267), Hall Tests (253), Intercept and Mail-back (487), Intercept and Web (198), Telephone Contacts (52), Home/Workplace Interviews (578), Online Panel (269), Mailing Lists/Social Media (80), Unsolicited Mail-out (70), Big Data (31), Other and Combinations (94), Not Specified (191)	Yes
Source of Evidence	As Table 2	Yes
World Bank Income Category	High Income (2667), Upper-Middle Income (153), Lower-Middle Income (52)	Yes
Main Aim of Study	To Estimate Value of Variable of Interest: Yes (929), No (1943)	No

recovered the same pattern of results. This similarity is not surprising, and indeed encouraging, given that *Wait*, *AccEg*, *OVT* and *Conn* might reasonably be regarded to have broadly similar levels of disutility to *Walk*. The same caveat applies regarding segmentation effects for the other attributes,¹⁹ but we note, as expected, lower multipliers for *Headway*, *Disp-earlier*, *Disp-later* and *Disp-nd*, larger multipliers for *MeanLate-Distⁿ*, *MeanLate-X/Y* and *SDL*, and broadly sensible absolute and relative effects for *Cong-nd*, *Cong-mod* and *Cong-hvy*.

In addition to the main effects distinguishing between the first 18 attributes set out in section 1.2, additional variables were specified to represent the remaining three main effects (19–21) of *SDE-withRR*, *SDL-withRR* and *RR-withSDE/L*. Initially, these entered the model alongside *SDE*, *SDL* and *RR* as follows:

$$\alpha_{16}SDE + \alpha_{19}SDEwithRR + \alpha_{17}SDL + \alpha_{20}SDLwithRR + \alpha_{18}RR + \alpha_{21}RRwithSDE/L \quad (3)$$

All terms are dummy variables. Hence where, say, *SDL* has been estimated without *RR* the effect is represented by α_{17} whereas it is $\alpha_{17} + \alpha_{20}$ when the multiplier is obtained from a model containing both *SDL* and *RR*. It turned out that the estimates of α_{19} , α_{20} and α_{21} were similar and hence combined into a single interaction denoted *TwoTerms* in Table 7. This indicates that where *SDE*, *SDL* and *RR* were estimated simultaneously in a given study the individual multiplier terms are 36% lower as a result of spreading their effects across additional terms. This could be because of correlation between parameter estimates in the underlying models but could also indicate that *SDE/SDL* and *RR* do not on their own fully represent travel time variability effects.

Study-specific fixed-effects were specified for the 483 studies that provided at least 2 observations, and hence the studies with just one observation serve as the base. An iterative process was followed with the removal of the least significant effects until all those remaining were significant at the 5% level.²⁰ This resulted in 127 fixed-effects, only 26% of those initially specified. The findings are reported as Model II, though for reasons of space the fixed-effects coefficient estimates themselves are not reported. Although many of the latter indicated quite large proportionate effects, taken overall the unweighted (weighted) average was only 0.032 (0.018). Whilst there was a tendency for the fixed-effects to have higher t-ratios for studies with more observations, as would be expected, no obvious

¹⁹ Section 5 considers implied multipliers across a range of circumstances covered by the segmentation variables.

²⁰ We retained this ‘high bar’ since we have no theoretical basis for retaining fixed effects that were not significant at the usual 5% level.

Table 7
Meta-models.

	Model I Segmented Model		Model II Segmented Model with Fixed-Effects		Model III Fixed-Effects Outliers Removed	
	Coeff (t)	%Effect	Coeff (t)	%Effect	Coeff (t)	%Effect
Constant	0.497 (18.1)		0.567 (21.9)		0.557 (25.5)	
Base Multiplier ($Walk$) = $e^{Constant+0.5MSE}$	1.85		1.91		1.84	
Main Effects						
<i>Walk</i>	<i>Base</i>		<i>Base</i>		<i>Base</i>	
<i>Wait</i>	—		—		—	
<i>AccEg</i>	—		—		—	
<i>OVT</i>	—		—		—	
<i>Conn</i>	—		—		—	
<i>Headway</i>	−1.097 (26.6)	−66.6%	−1.075 (30.8)	−65.9%	−1.029 (34.5)	−64.3%
<i>Search</i>	−0.277 (3.0)	−24.2%	−0.291 (3.7)	−25.2%	−0.271 (4.1)	−23.7%
<i>Cong-nd</i>	−0.303 (5.3)	−26.1%	−0.338 (6.9)	−28.7%	−0.329 (8.0)	−28.0%
<i>Cong-mod</i>	−0.395 (5.6)	−32.6%	−0.377 (6.2)	−31.4%	−0.363 (7.2)	−30.4%
<i>Cong-hvy</i>	—		—		—	
<i>Disp-earlier</i>	−1.401 (20.2)	−75.4%	−1.911 (28.1)	−85.2%	−1.852 (32.1)	−84.3%
<i>Disp-later</i>	−1.171 (16.8)	−69.0%	−1.660 (24.7)	−81.0%	−1.627 (28.1)	−80.3%
<i>Disp-nd</i>	−1.277 (10.3)	−72.1%	−1.812 (15.1)	−83.7%	−1.875 (18.2)	−84.7%
<i>MeanLate-Distⁿ</i>	0.409 (3.2)	+50.5%	0.339 (3.2)	+40.4%	0.349 (3.9)	+41.8%
<i>MeanLate-X/Y</i>	0.467 (7.0)	+59.5%	0.431 (6.7)	+53.9%	0.389 (7.2)	+47.6%
<i>SDE</i>	−0.859 (10.1)	−57.6%	−0.804 (10.2)	−55.2%	−0.772 (10.5)	−53.8%
<i>SDL</i>	0.264 (3.4)	+30.2%	0.225 (3.0)	+25.2%	0.264 (4.1)	+30.2%
<i>RR</i>	−0.917 (10.1)	−60.0%	−0.736 (9.5)	−52.1%	−0.595 (8.6)	−44.8%
<i>TwoTerms</i>	−0.451 (4.9)	−36.3%	−0.586 (7.0)	−44.3%	−0.505 (6.7)	−39.6%
Segmentation Effects						
<i>Mode Used Segmentations</i>						
<i>WWAOC_UseTrainLRTMetro</i> (162/160)	−0.138 (3.4)	−12.9%	−0.126 (3.5)	−11.8%	−0.131 (4.4)	−12.3%
<i>Headway_UseCar</i> (107/97)	0.147 (2.7)	+15.8%	—		—	
<i>Headway_UseLRTMetro</i> (20/18)	0.494 (4.4)	+63.8%	0.373 (4.0)	+45.2%	0.347 (4.2)	+41.5%
<i>Headway_UseAir</i> (9/8)	−0.545 (3.2)	−42.0%	−0.493 (3.5)	−38.9%	−0.719 (5.8)	−51.3%
<i>Disp_UseCar</i> (121/115)	0.175 (2.6)	+19.1%	0.376 (5.5)	+45.6%	0.413 (7.1)	+51.1%
<i>Late_UseTrainPT</i> (71/65)	0.271 (3.2)	+31.1%	0.112 (1.6)	+11.9%	0.141 (2.3)	+15.1%
<i>RR_UseCar</i> (83/72)	0.148 (1.6)	+16.0%	—		—	
<i>Journey Purpose Segmentations</i>						
<i>Disp_EB</i> (33/32)	0.216 (2.3)	+24.1%	0.261 (3.3)	+29.9%	0.166 (2.4)	+18.1%
<i>RR_Commute</i> (73/67)	0.206 (2.5)	+22.8%	0.158 (2.2)	+17.1%	0.080 (1.2)	+8.3%
<i>Spatial Segmentations</i>						
<i>Headway_Inter</i> (148/134)	−0.283 (5.7)	−24.7%	−0.355 (7.7)	−29.9%	−0.344 (8.6)	−29.1%
<i>Rely_Inter</i> (48/43)	−0.149 (1.7)	−13.8%	−0.212 (2.5)	−19.1%	−0.221 (2.8)	−19.8%
<i>Rely_Mixture</i> (60/52)	−0.315 (3.8)	−27.0%	−0.135 (1.9)	−12.6%	−0.213 (3.4)	−19.2%
<i>CongSearch_NonMetropolis</i> (288/285)	0.186 (3.5)	+20.4%	0.142 (3.0)	+15.3%	0.130 (3.3)	+13.9%
<i>Headway_UMLM</i> (17/17)	−0.227 (1.9)	−20.3%	−0.428 (4.0)	−34.8%	−0.466 (5.3)	−37.3%
<i>Data Type Segmentations</i>						
<i>WAC_RP</i> (142/136)	0.115 (2.6)	+12.2%	0.145 (3.7)	+15.6%	0.165 (5.0)	+17.9%
<i>Wait_RP</i> (78/72)	0.242 (4.1)	+27.4%	0.241 (4.8)	+27.3%	0.289 (6.7)	+33.5%
<i>OVT_RP</i> (33/31)	0.407 (4.6)	+50.2%	0.104 (1.7)	+11.0%	0.190 (2.4)	+20.9%
<i>Headway_RP</i> (21/18)	−0.298 (2.7)	−25.8%	−0.269 (2.7)	−23.6%	−0.260 (2.9)	−22.9%
<i>Rely_RP</i> (19/17)	0.196 (1.7)	+21.7%	0.219 (2.0)	+24.5%	0.151 (1.6)	+16.3%
<i>Intertemporal Segmentations</i>						
<i>Walk_1991+</i> (428/421)	−0.083 (2.4)	−8.0%	−0.159 (5.0)	−14.7%	−0.153 (5.7)	−14.2%
<i>Wait_2001+</i> (282/273)	−0.113 (2.9)	−10.7%	−0.174 (4.9)	−16.0%	−0.166 (5.5)	−15.3%
<i>ConnOVT_2001-10</i> (51/50)	−0.101 (1.4)	−9.6%	−0.123 (1.6)	−11.6%	−0.103 (1.6)	−9.8%
<i>ConnOVT_2011+</i> (53/51)	−0.262 (3.6)	−23.0%	−0.187 (3.0)	−17.1%	−0.159 (3.1)	−14.7%
<i>AccEg_1991+</i> (259/243)	−0.099 (2.5)	−9.4%	−0.167 (4.5)	−15.4%	−0.161 (5.1)	−14.9%
<i>RR_2011+</i> (77/70)	−0.190 (2.4)	−17.3%	−0.325 (3.8)	−27.7%	−0.412 (5.5)	−33.8%
RSS	672.60		436.42		283.50	
MSE	0.237		0.161		0.110	
$\bar{R}^2$	0.544		0.690		0.756	
Observations	2872		2872		2731	

Note: The number of observations for the segmentation terms is provided in parentheses alongside the variable name. The first figure relates to Models I and II and the second to Model III.

pattern to the fixed-effects was apparent on inspection.

The inclusion of the fixed-effects has little impact on the main effect estimates in Table 7. There is though more of an impact on the segmentation terms, with *Headway_UseCar* and *RR_UseCar* becoming far from significant, and hence removed, and some large variations, notably for the %Effect of *Disp_UseCar*, *Late_UseTrainPT*, *RelyMixture*, *Headway_UMLM* and *OVT_RP*. We might expect some effects previously attributed to the segmentation terms to be discerned by study-specific fixed-effects but for most variables the variation in %Effect is relatively minor with an almost equal split between the %Effect becoming larger or smaller in absolute. As would be expected with the addition of 127 statistically significant new terms, the $\bar{R}^2$ improves somewhat from 0.544 to 0.690.

Model III then removes 'outlier' observations with standardised residuals outside the range ± 1.96 which can be taken to represent the 5% of observations which are of poorest quality or remain difficult to explain. This process, as is entirely to be expected, increases the $\bar{R}^2$, with a notable improvement from 0.690 to 0.756. One fixed-effect became insignificant and was removed. Compared to Model II, the %Effects for the main effect terms vary little and almost always have larger t ratios, whilst for the segmentation terms the %Effect is generally little different, with the possible exceptions of *RR_Commute* and *OVT_RP*, with mainly larger t ratios and an almost equal balance of %Effect becoming larger or smaller in absolute. The unweighted mean of the significant fixed-effects was only 0.041, with the weighted mean zero to three decimal places. Weighting by the number of observations in each study would imply a revised constant for Model III of 0.562 which is hardly different to the estimated constant of 0.557 and hence would not materially impact the multipliers implied by the model according to whether the detected fixed-effects were regarded to be genuine or not.

Our preference is for Model III that incorporates significant study-specific effects and removes what we deem to be outlier observations, and it is the findings of this model that we now discuss.

4.1. Mode effects

Information was collected on both the mode or modes used and the mode or modes being valued, as indicated in Table 6. It turned out that the only significant segmentations related to mode used.

Initial models indicated a tendency for the multipliers related to the largely public transport specific variables to be similarly lower for public transport users, although the effects for individual attributes were not significant. Given that public transport users will be more accustomed to these attributes, and self-selection means that they would value them less, we specified a single segmentation effect for *Walk*, *Wait*, *AccEg*, *OVT* and *Conn* and a significant effect was obtained for train, LRT and metro users (*WWAOC_UseTrainLRTMetro*) which indicated a multiplier 12.3% lower. Whilst monetary values of *WWAOC* might be conjectured to vary by more than this when compared with, say, car users, the higher values of *IVT* for car users could have had a dampening effect on the variation.

Car users can be expected to dislike the inability to be able to travel when they want to relatively highly and hence to have relatively large headway values. An effect (*Headway_UseCar*) was apparent in Model I, denoting 15.8% larger multipliers, but it was not significant in Model III, and again the relatively large valuation of *IVT* for car users could have had a bearing.

The effect apparent on headway multipliers for LRT and Metro users (*Headway_UseLRTMetro*) of 41.5% is one of the largest segmentation effects recovered. This is attributed to LRT and Metro users being accustomed to very high service frequency and that the attraction in using such systems is critically dependent upon it.

Although the number of headway multiplier observations for air travellers (*Headway_UseAir*) is low, the highly significant effect is the largest in the model at -51.3%. Air travel is not characterised by high frequencies, and it is generally not expected, and it is reasonable that air users with their less frequent and highly planned journeys deem headway to be somewhat less important than other mode users.

Car users have 51.1% larger multipliers for displacement time (*Disp_UseCar*) which presumably reflects their strong preference for being able to travel at their desired departure time.

A reliability effect was recovered for train and PT (bus, metro and LRT) users (*Late_UseTrainPT*). This was based on the composite *Late* term since separate effects for *MeanLate-Distⁿ*, *MeanLate-X/Y* and *SDL* were not significant but were indicating an effect was present. A 15.1% larger *Late* multiplier was apparent for these users. This may stem from a greater awareness of late running for modes with schedules and lesser mitigation opportunities for train and PT users.

4.2. Journey purpose

It is expected, and commonly found, that the monetary valuation of *IVT* is largest for employers' business (EB) trips, followed by commuting and leisure, all else equal. The effect on multipliers will depend upon the extent to which journey purpose impacts the monetary values of the non-*IVT* variables similarly. The Cost Savings Approach to valuing business travel time savings would contend that all forms of travel time are unproductive and savings in any of them would be converted into productive use to be valued at the marginal product of labour (Harrison, 1974). This approach is widely used in official recommendations around the world (Mackie et al., 2014) and assigns the same weight to *Walk* and *Wait* as to *IVT* for business travel.

Despite extensive searching for journey purpose effects, only two significant segmentation effects could be estimated. Business travellers have larger multipliers for displacement time (*Disp_EB*), and it is understandable that they are more averse to switching from their desired departure time, whilst unsurprisingly commuters were found to have larger *RR* multipliers (*RR_Commute*) although only 8.3% larger and we were unable to detect an equivalent effect for the other late time attributes. There was no support for the property of the Cost Savings Approach of valuing *Walk*, *Wait* and other non-*IVT* time variables the same as *IVT* for business travellers.

4.3. Distance and spatial effects

The analysis of distance and spatial effects covered: whether the journey was urban, inter-urban or was a mixture of the two; categories of urban hierarchy, essentially distinguishing between the size of urban locations; and World Bank country income classifications, which cover high (H), upper-middle (UM) and lower-middle (LM) income definitions. Limited country specific effects were also investigated.

The headway multiplier was found to be 29.1% lower for inter-urban trips (*Headway_Inter*) which is not surprising since service frequency is much more of an issue for travellers making shorter and more regularly made journeys and high frequencies are not expected on longer distance journeys. It might also reflect the greater composition of wait time for shorter distance trips, where frequencies are higher, as discussed in section 3.3.

The composite *Rely* multiplier, defined above, was specified given relatively small sample sizes. It was found to be 19.8% lower than for urban trips (*Rely_Inter*). Lateness is more common and less unacceptable for longer distance trips. Where no distinction was made between urban and inter-urban trips (*Rely_Mixture*), the effect was essentially the same.

An argument sometimes found in the literature relates to a decrease in available time budget for longer trips, with less time remaining for other activities and making travel time savings for longer trips more valuable. This would not impact the time-related multipliers, but could contribute to these lower multipliers for *Headway* and *Rely* for inter-urban trips.

For urban trips, a further distinction was made according to whether the location was a major metropolis, including capital cities, a metropolitan area, an urban area, a rural area, or some mix of these area types. The key effects that emerged were that the congestion and search time multipliers were higher outside of the major metropolises. A combined term for these attributes (*CongSearchNonMetropolis*) indicated 13.9% higher valuations, which is interpreted as those in major metropolises being resigned to congestion and difficulties finding a parking space and hence are relatively less concerned about it. In addition, those who do not like congested travel conditions might be less likely to live in major metropolises.

The remaining statistically significant spatial effect relates to the World Bank income classification. *Headway* was found to have a 37.3% lower multiplier in UM and LM income countries (*Headway_UMLM*). This lower valuation is not surprising. Whilst it is encouraging from the perspective of transferability that there are not widespread differences according to income classification, the limited number of observations for UM and LM income countries should be borne in mind.

As a final test, we explored whether the composite multipliers of *WWAOC*, *Disp*, *CongSearch*, and *Rely* were different for the UK, Australia and the Netherlands which are the three countries with very large numbers of multiplier observations. No significant effects were apparent.

4.4. Data type

Whilst there is primary and review evidence that monetary valuations differ according to whether they are estimated to RP or SP data (Wardman, 1988; Brownstone and Small, 2005; Shires and de Jong, 2009; Wardman et al., 2025), there has been little detailed analysis of whether time multipliers exhibit any such variation. Some prior cross-tabulations indicated that multipliers obtained from SP evidence were lower, and the differences were not trivial.

The dataset contains 342 (12%) RP observations and in 202 (59%) cases the journey components were obtained from network data and models generally estimated to secondary choice data with the remaining 140 (41%) based on reported data and models estimated to purpose collected survey data. Unsurprisingly, RP data is more common in earlier studies, providing 118 (33%) of observations up to 1990, and in the subsequent periods 1991–2000, 2001–2010 and after 2010 the respective figures are 65 (8%), 113 (10%) and 46 (7%). RP multipliers were available for all attributes except *MeanLate-Distⁿ* but with only 2 available for *SDE*. As can be seen from Table 7, *WWAOC* terms form the most RP observations, with around three-quarters, made up of 70 for *Walk*, 78 for *Wait*, 59 for *AccEg*, 33 for *OVT* and 13 for *Conn*. *CongSearch* provided 8% with around 6% each for *Headway*, *Disp* and *Rely*. The main travel context covered was mode choice (68%), followed by route choice (14%), mode and destination choice (7%), departure time choice (6%) with other contexts and some combinations making up the remaining few, whilst 90% were drawn from developed countries.

Of the remaining observations, 133 were obtained from joint RP-SP models and 2397 from some form of SP model. SP evidence was treated as the base and, unsurprisingly given that SP data dominates RP-SP models but also perhaps because of the relatively few observations, there were no significant effects relating to RP-SP evidence.

An important issue to discuss here is the specification of reliability terms in RP models, given that it can be regarded to be more challenging than for SP models where the degree of lateness or spread of journey times is explicitly presented, and this might cast doubts on the validity of RP-based reliability multipliers. The RP reliability evidence here is drawn from 11 studies covering an even mix of route choice, mode choice and departure time choice studies, several of which paid particular attention to the measurement of reliability. Reliability was estimated by a range of methods, including the use of GPS, Smart data, traffic detectors and journey time surveys, along with respondents' reported travel times and/or lateness. Even if these measures do not always accurately represent travellers' perceptions, and the quality of the reliability estimates is better in some studies than others, these effects might offset in our meta-model.

Model I contains five segmentation terms for multipliers obtained from RP data. The effects for *Walk*, *AccEg* and *Conn* when estimated separately were indicating larger effects for RP data which were very similar but none were individually significant. They were therefore combined into a single term (*WAC_RP*). The coefficient estimate is highly significant and indicates that these multipliers are 17.9% larger than when obtained from SP data. The *Wait* multiplier when RP based (*Wait_RP*) is 33.5% larger whilst that for *OVT* (*OVT_RP*) is similar to *WAC_RP* at 20.9% larger. The impact on the *Rely* composite term (*Rely_RP*) is not significant at the usual 5% level

but its 16.3% effect is in line with other effects. The one exception to the positive RP effect is for *Headway* (*Headway_RP*) which is found to have a 22.9% lower value.

These are interesting and important findings. They could indicate that in SP studies respondents focus primarily on cost and IVT as the most important factors and the artificial and sometimes complex nature of the task results in other attributes not being fully considered which therefore leads to lower valuations than otherwise. Such an effect might be exacerbated by unrealistic attribute variation or poor explanation in SP studies, such as varying walk times to a fixed destination or varying public transport wait times when the traveller always turns up for the service X minutes before its scheduled departure. On the other hand, RP data requires assumptions about the trade-offs actually made by passengers, leading to less certainty regarding the reliability of inferred values, whilst RP multipliers might be larger due to possible confounding effects given that 33% of the multiplier evidence up to 1990 was from RP studies yet in subsequent periods the proportion is lower at around 10%.²¹ As for *Headway*, where the RP values are lower, a contributory factor could be poor specification of service headway in RP models²² or the use of service headway when in fact it is wait time that underpins choices.

4.5. Intertemporal effects

Whilst specific studies can provide evidence on how multipliers vary with mode, distance or even data type, they provide no insight into intertemporal effects, and this is where meta-analysis is particularly valuable.

Segmentations were undertaken according to when the data used in estimation was collected, with up to 1990 serving as the base and then segmentations specified for 1991–2000, 2001–2010, and after 2010, the exception being for RR where the base period was up to 2000. There was insufficient data to isolate post-covid variations.

Significantly different segmentation effects specified for the different time periods were not apparent and hence categories have been pooled according to similar effects and goodness of fit. Whilst an element of judgement is here involved, there is clearly support for several multipliers falling over time.

Model III finds the *Walk* multiplier (*Walk_1991+*) to be 14.2% lower post 1990 with similar effects for *Wait* (*Wait_2001+*) of a 15.3% reduction post 2000 and for the *AccEg* multiplier (*AccEg_1991+*) which is 14.9% lower post 1990. The effects upon *Conn* and *OVT*, which were combined on the grounds of sample size, distinguish the years 2001–2010 (*ConnOVT_2001-10*) and post 2010 (*ConnOVT_2011+*) since, even though the former is significant only at the 10% level, there seems to be an effect at work here. The former indicates a 9.8% reduction and the latter a 14.7% reduction. As for RR, it was found to be 33.8% lower post 2010 (*RR_2011+*), the largest intertemporal effect detected, although no significant effects could be discerned for the other late time variables.

There is evidence that the multipliers have fallen over time. The reason for the reductions in the *Walk*, *Wait*, *Conn* and *OVT*, and *AccEg* multipliers is likely that advances in the quality and quantity of mobile information and communications technology (ICT) means that more and better worthwhile activities can be undertaken while walking and waiting. However, the effects are not large, and this may be because these effects on the disutility of travel time also apply to the denominator term of IVT although not to the same extent. The impact on RR could be attributed to a greater acceptance of unreliability if there is a perception of an inevitably more unreliable world, as well as more and better real-time information which supports mitigation.

4.6. Data quality issues

A number of data quality issues were examined which merit pointing out. There were no significant impacts detected on the multipliers from: the use of advanced modelling techniques, such as mixed logit and hybrid choice models; efficient SP design procedures; whether the evidence was drawn from peer-reviewed academic journals, or indeed national studies undertaken to inform official recommendations; or the sample sizes of the models that provided the multipliers. But then why should poorer or higher quality studies yield systematically different multipliers across the data assembled? A random effect overall is by no means implausible and this is supported by the study-specific fixed-effects averaging close to zero. As for the precision of our estimates and quality, the removal of outlier observations did tend to increase precision but weighting the multiplier observations in Eq. (2) by the sample sizes of the models that provided those multipliers had very little impact.

5. Application: implied multipliers

Whilst meta-analysis can provide useful methodological insights, a primary motivation for this research as set out in section 1.1 is to summarise the evidence so as to be in a position to provide recommendations and also to make comparisons with official recommendations, where they exist, and long-established conventions. Section 1.1 also specified that an attraction of meta-analysis is to quantify variations in multipliers, not least to aid transferability across contexts. Table 8 provides illustrative multipliers for a range of 46 scenarios, labelled as S1–S46, selected to cover combinations of variables upon which recommendations should be based and to demonstrate the range of potential variation across circumstances. The characteristics of the scenarios differ across multipliers given

²¹ The more recent dominance of SP methods can be attributed to estimation efficiency and cost advantages, although it is not unknown, particularly in major or challenging studies, that corroboration against suitable RP evidence is deemed desirable.

²² Although there are also approximations involved in the measurement of other attributes in RP models, notably for public transport where significant fare differentiation exists and where the time involved in accessing services exhibits considerable inter-personal variation.

that the variables driving variations in their valuations differ. The implied multipliers for scenarios S1–S46 are based on Model III of Table 7.

Given that the estimated meta-model involves a logarithmic transformation, the unbiased predictor of the multiplier involves adding half the mean square error (MSE) to the estimate of the constant term (Miller, 1984). The revised constant used in deriving the implied multipliers is therefore 0.612. The fixed-effects have been ignored but, as already pointed out, their impact on the constant would be very slight.

The multipliers covered in Table 8 are the most important ones, in line with the official recommendations covered in Table 1. The scenarios are selected to cover the vast majority of the factors that drive the variations within each multiplier, with an emphasis on the range in which the implied multipliers might lie; the intertemporal effects and differences between RP and SP based multipliers are of particular interest. Hence the focus here is upon comparing the implied multipliers across a row rather than inspecting variations within columns.

The primary modes, prior to any combinations in our analysis, are car, bus, train, LRT/Metro, taxi, motorcycle, ferry and air. The multipliers were found to vary with mode used but there were no additional effects attributed to the mode to which the valuation relates.²³ The scenarios in Table 8 distinguish all the modal effects identified for each multiplier and also all other segmentation effects and their base levels except for *Rely Mixture*, given that it is more appropriate to distinguish urban and inter-urban trips, and *Headway UMLM*, where the S17–S24 multipliers relate to high income countries and would be 37% lower otherwise.

Whilst the variations in multipliers arising from specific influential variables can be relatively small, there are some large but nonetheless credible variations when adding up over various effects as is apparent across the scenarios specified in Table 8. The implied multipliers for an attribute tend to vary, as would be expected, around the reference multipliers provided in the second column, which are the mean values across all observations as reported in Table 5.

Implied multipliers are provided for the two key variables of *Walk* and *Wait*. The convention of valuing these at twice IVT, now over 50-years old, is still used today in many contexts, including appraisal guidelines as set out in Table 1. If the multipliers are based upon older evidence and RP data (S5, S6, S13 and S14), which would have been the basis of the convention, the historic multiplier of 2 for *Walk* and *Wait* would indeed seem justified. Some commentators have questioned whether *Walk* and *Wait* should still be valued at twice IVT, with Table 1 showing that several countries recommend values below 2, and our results offer some support for lower multipliers.

Walk and *Wait* multipliers have gone down over time, presumably due to the increased opportunities to use time while walking or waiting and to the improvements in safety and walking/waiting conditions more widely. Retaining the RP approach as preferred but allowing for the reduction in multipliers over time (S1, S2, S9 and S10), there would seem to be justification for lower *Walk* multipliers, say around 1.75, but not for lower *Wait* multipliers. If, however, the results of the more widely used SP methods are preferred and combined with the more recent years (S3, S4, S11 and S12) then there would be strong support for moving to a figure around 1.5.

As for *Headway*, it is the one attribute where the SP data provides larger multipliers. In this context, it might be claimed that the SP evidence is more robust since the specification of service headway in such models will be more accurate than in RP models. However, a limitation of our results is that the unit *Headway* valuation might be expected to vary with the level of service frequency, with travellers valuing a given improvement to an hourly service more than to a service every half hour or 15 min, but the literature often does not examine such non-linearities and we have no direct empirical evidence on it. Moreover, some of the official recommendations in Table 1 exhibit a diminishing multiplier as the service headway increases due to moving from random arrivals and wait time to planned arrivals and headway penalties. For this reason, the *Headway* multipliers here might best serve a benchmarking purpose, with the lower multipliers for inter-urban trips (S19 and S22) being useful in this regard, and as such they provide an encouraging degree of support for central recommendations of the official guidance set out in Table 1.

Car journey time spent in congested conditions is not routinely considered in transport appraisal. However, there is now a large and convincing body of evidence that a premium is attached to such time. No intertemporal or data type effects were detected. For *Cong-nd*, the multiplier is 1.33 within major metropolises (S25) and 1.51 elsewhere (S26). Distinguishing *Cong-mod* and *Cong-hvy* provides credible variations around these figures, with moderate and heavy congestion multipliers of 1.28 and 1.84 for major metropolises and 1.46 and 2.10 elsewhere. The official guidance for the congestion multiplier in Table 1 is supported by the findings here, albeit with the figures for Singapore for light and heavy congestion on the low side.

The other attribute for which there are no intertemporal or data type effects is displacement time. Here there is relatively little variation in implied multipliers, with a figure in the range 0.3 to 0.4 relevant to most travellers and very little difference between earlier and later departures. These tend to support the only official recommendations, of which we are aware, for Australia.

The remaining attributes relate to travel time variability. This is a contentious issue and is likely to lead to biased responses in SP exercises, and our dataset contains a few very large multipliers for *MeanLate-Dist*ⁿ and *MeanLate-X/Y*. However, the RP based evidence implies even larger multipliers. As a result, it is unclear which set of numbers should be preferred, but what is clear is that the multipliers are large and that it cannot be argued that they are large simply because they are based on SP evidence. Taken overall, the evidence broadly supports official recommendations, but lower multipliers for inter-urban trips and higher multipliers for train and public transport users not apparent in the official recommendations would seem to be warranted.

As for the *RR* multipliers, they are lower for more recent data, presumably on the grounds that unreliability has become more expected and accepted, and these figures are preferred. They are also lower for inter-urban trips. The variation in the implied *RR*

²³ If multipliers are required which do not distinguish by mode used, then weighted averages can be taken.

Table 8
Illustrative Implied Multipliers

<i>Walk</i>		S1	S2	S3	S4	S5	S6	S7	S8
	Reference Multiplier	Train/LRT/Metro, RP, 1991+	Car/Bus/Air/Other, RP, 1991+	Train/LRT/Metro, SP, 1991+	Car/Bus/Air/Other, SP, 1991+	Train/LRT/Metro, RP, ≤1990	Car/Bus/Air/Other, RP, ≤1990	Train/LRT/Metro, SP, ≤1990	Car/Bus/Air/Other, SP, ≤1990
	1.67	1.64	1.87	1.39	1.58	1.91	2.17	1.62	1.84
<i>Wait</i>		S9	S10	S11	S12	S13	S14	S15	S16
	Reference Multiplier	Train/LRT/Metro, RP, 2001+	Car/Bus/Air/Other, RP, 2001+	Train/LRT/Metro, SP, 2001+	Car/Bus/Air/Other, SP, 2001+	Train/LRT/Metro, RP, ≤2000	Car/Bus/Air/Other, RP, ≤2000	Train/LRT/Metro, SP, ≤2000	Car/Bus/Air/Other, SP, ≤2000
	1.72	1.83	2.09	1.37	1.56	2.16	2.46	1.62	1.84
<i>Headway</i>		S17	S18	S19	S20	S21	S22	S23	S24
	Reference Multiplier	Train/Bus/Car/Other, Urban, RP	LRT/Metro, Urban, RP	Train/Car, Inter, RP	Train/Bus/Car/Other, Urban, SP	LRT/Metro, Urban, SP	Train/Car, Inter, SP	Air, RP	Air, SP
	0.60	0.51	0.72	0.36	0.66	0.93	0.47	0.18	0.23
<i>Cong-nd</i>		S25	S26						
	Reference Multiplier	Metropolis	Non-Metropolis						
	1.48	1.33	1.51						
<i>Disp-nd</i>		S27	S28	S29	S30				
	Reference Multiplier	Car, EB	Car Non-EB	Non-Car EB	Non-Car Non-EB				
	0.58	0.50	0.43	0.33	0.28				
<i>MeanLate-Dist^a</i>		S31	S32	S33	S34	S35	S36	S37	S38
	Reference Multiplier	Train/PT, RP, Urban	Train/PT, RP, Inter	Car/Other, RP, Urban	Car/Air/Other, RP, Inter	Train/PT, SP, Urban	Train/PT, SP, Inter	Car/Other, SP, Urban	Car/Air/Other, SP, Inter
	2.53	3.50	2.81	3.04	2.44	3.01	2.41	2.61	2.10
<i>RR</i>		S39	S40	S41	S42	S43	S44	S45	S46
	Reference Multiplier	Comm, RP, 2011+, Urban	Non-Comm, RP, 2011+, Urban	Non-Comm, RP, 2011+, Inter	Comm, SP, 2011+, Urban	Non-Comm, SP, 2011+, Urban	Non-Comm, SP, 2011+, Inter	Comm, RP, ≤ 2010, Urban	Non-Comm, SP, ≤2010, Inter
	0.93	0.85	0.78	0.63	0.73	0.67	0.54	1.28	0.82

Note: All mode terms relate to users. Comm denotes commuting and EB denotes employer's business. Other covers the primary modes of ferry, motorcycle and taxi.

multipliers based on more recent data (S39-S44) is somewhat less than the very wide range of official *RR* multipliers in Table 1 between 0.1 and 1.4. A range between 0.6 and 0.9 would seem more appropriate, with additional allowance for lower multipliers for inter-urban trips.

6. Conclusions

This study has reported the most extensive review and meta-analysis of time multipliers ever made, spanning many years and countries. A wide range of important transport planning and appraisal-relevant multipliers has been covered: walk, wait, access/egress, out-of-vehicle and connection time; public transport service headway; parking search time; time spent in congested traffic, with distinctions according to moderate and heavy congestion; travel time variability, covering lateness relative to timetable or expectation, schedule delay early, schedule delay late and the standard deviation of travel time (reliability ratio); and displacement time, including earlier and later departure time shifts.

The mean multipliers are credible and, where relationships exist between them, the results are broadly consistent. Walk, wait, access/egress, out-of-vehicle and connection time tend to have similar multipliers, which seems reasonable, whilst the premium attached to congested time seems sensible and the ordering of the moderate congestion, congestion and heavy congestion multipliers is consistent. Late arrival time is particularly highly valued with headway and displacement time having multipliers less than one as would be expected.

The impact of a wide range of variables on the multipliers has been investigated. Of particular interest are the findings that many multipliers have fallen over time and many multipliers depend upon the data used in estimation.

Estimated reductions in the walk, wait, access/egress, out-of-vehicle and connection time multipliers over time are attributed to advances in the quantity and quality of ICT which transform what was previously largely dead time into time where worthwhile activities can be undertaken. The reduction in the reliability ratio over time presumably stems from greater acceptance of unreliability in what might be perceived to be an increasingly unreliable transport world, along with greater opportunities for mitigation due to the increased availability of real-time information.

As for data type, RP based multipliers are larger for walk, wait, access/egress, out-of-vehicle and connection time and also a composite term covering mean lateness, schedule delay late and the reliability ratio. With regard to the various travel time terms that are not in-vehicle time, it is convincing that all five were found to have larger multipliers when obtained from RP data. This could be attributed to the artificial and sometimes complex nature of SP exercises and respondents' primary focus may have been on cost and *IVT* which will lead to lower multipliers for other attributes to the extent that they might not have been given as much attention. These effects might be exacerbated by unrealistic attribute variation or poor explanation of that variation in SP exercises. It is, however, unclear why the RP multiplier for the composite late time is larger. But on the other hand, SP methods are now the default for estimating valuations of travel time attributes.

The exception to the above pattern of results was for the headway multiplier where SP based valuations are larger and this could stem from the poor specification of service headway in RP models or the use of headway when in fact it is wait time that underpins choices.

A few other influences were apparent from mode used, journey purpose and distance. Perhaps reassuringly, the only effect from World Bank country income classification was for service headway and we did not discern any country specific effects. The investigation of quality related aspects of observations and studies did not raise any concerning issues.

The number of modifiers estimated to the multipliers is limited. A contributory factor is that some segmentation variables might have impacted the numerator and denominator elements of the multiplier terms similarly and hence will tend to cancel out or make it more difficult to recover significant effects. Indeed, we would expect far less variation in time multipliers than in the monetary valuations of the attributes here covered. Whilst the impacts of specific modifier variables tend to be relatively minor, quite large variations in multipliers can be apparent as a result of combining various individual modifier effects when, as reported in Table 8, the meta-model is applied to estimate implied multipliers across a range of scenarios.

The estimated meta-model implies plausible multipliers and can therefore serve a useful purpose in providing multipliers where none exist or benchmarking existing or emerging evidence. When compared to the official multiplier guidance that we have identified, the following conclusions can be drawn:

- There has been speculation that the long-established convention of valuing walk and wait time at twice *IVT* may no longer be valid, and the results presented here would provide some support based on detected reductions in walk and wait multipliers over time. A figure of 1.75 for the walk time multiplier would seem justified if the RP basis of the conventional wisdom was retained. However, if SP results are regarded to be more reliable then a reduction to 1.50 for both walk and wait time multipliers is warranted, supporting the multipliers of 1.50 that are recommended by some administrations.
- The headway multipliers implied by the meta-model serve a useful benchmarking purpose and broadly support those recommended in official guidance.
- The displacement time multipliers could be used to significantly enhance existing official guidance which only exists for Australia.
- Official congestion multiplier guidance is generally supported by the findings here, but we would recommend adoption of the implied multipliers and particularly the distinction between moderate and heavy congestion.
- The results broadly support the large official late time multipliers, with additional distinctions between urban and inter-urban trips and between rail and public transport users and users of other modes warranted.

- Official guidance exhibits a very large range of reliability ratio multipliers, and the extreme valuations would seem to be difficult to justify. It is recommended that the meta-model's results are adopted; a range of around 0.6 to 0.9 would seem appropriate and within that lower valuations for inter-urban trips would seem sensible.
- The results do not support the widely used Cost Savings Approach to valuing all aspects of time the same for business travellers. This finding might challenge the legitimacy of the Cost Savings Approach.

CRedit authorship contribution statement

Mark Wardman: Writing – review & editing, Writing – original draft, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Paul Murphy:** Writing – review & editing, Methodology, Investigation, Formal analysis. **Manuel Ojeda-Cabral:** Writing – review & editing, Writing – original draft, Data curation. **Alex Mitrani:** Writing – review & editing, Investigation, Data curation. **Kazuyuki Neki:** Data curation. **Richard Martin Humphreys:** Resources, Funding acquisition, Data curation.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.tra.2026.104925>.

Data availability

Much of the data has been assembled largely by one of the authors without funding and is of considerable personal value

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